Comparison of Near-Infrared, Infrared, and Raman Spectroscopy for the Analysis of Heavy Petroleum Products

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Near-infrared (NIR) spectroscopy has been successfully applied to the determination of API (American Petroleum Institute) gravity of atmospheric residue (AR), which is the heaviest fraction in crude oil. This fraction is completely dark and very viscous. Preliminary studies involving Raman and infrared (IR) spectroscopies were also evaluated along with NIR spectroscopy. The Raman spectrum of AR was completely dominated by strong fluorescence from polycyclic aromatic hydrocarbons, called asphaltenes. IR spectroscopy provided reasonable spectral features; however, its spectral reproducibility was poorer and noisier than that of NIR. Although absorption bands in the NIR region were broad and less characterized, NIR provided better spectral reproducibility with higher signal-to-noise ratio (which is one of the most important parameters in quantitative calibration in comparison to Raman and IR spectroscopies). Partial least-squares (PLS) regression was utilized to develop calibration models. NIR spectra of AR samples were broad, and baselines were varying due to the strong absorption in the visible range. However, the necessary information was successfully extracted and correlated to the reference API gravity with the use of PLS regression. API gravities in the prediction set were accurately predicted with an SEP (standard error of prediction) of 0.22. Additionally NIR showed approximately three times better repeatability compared to the ASTM reference method, which directly influences the process control performance.

Index Headings: **Near-infrared; Infrared; Raman; Heavy petroleum products; API gravity; Partial least-squares.**

INTRODUCTION

Crude oil is a highly complex mixture of hydrocarbons and heteroatomic organic compounds of varying molecular weight and polarity. Crude oil is converted into many different products by a combination of physical and chemical processes known as refining. The first operation in refining is fractional distillation of crude oil.[1,2] This process splits crude oil into several fractions with different boiling ranges such as gases, straight-run gasoline, naphtha, kerosine, gas oil, and atmospheric residue (AR). From gases to AR, the boiling temperature range, the number of carbon atoms (molecular weight), and the complexity of composition increase. Straight-run gasoline, naphtha, kerosine, and gas oil (boiling temperature ranges from 30 to approximately 350 °C) are clear liquids and are easily measurable by vibrational spectroscopic methods without tedious and time-consuming sample pretreatments. Therefore, many vibrational spectroscopic techniques such as near-infrared (NIR), infrared (IR), and Raman spectroscopy combined with multivariate calibration methods have been widely applied to the determination of physical and chemical properties of petroleum and petrochemical products.[3–10] Multivariate calibration methods[11,12] are necessary to extract analytical information, since these samples are highly complex in terms of composition, and resulting spectral bands are highly overlapped. The applications of vibrational techniques in this field have gradually increased to replace conventional analyses in order to shorten the analysis time and improve the precision, which are related to the performance of process control and optimization.

AR samples (boiling temperature of approximately 350 to 800 °C are used as the main fuel for power stations, ships, and large heating installations) are completely dark in color and very viscous (solid at room temperature), and are hard to handle. Therefore, application of vibrational techniques to AR has not been easy in comparison to lighter products that are clear and transparent liquids. Moreover, AR has solid suspensions and particles that adversely affect the reproducibility of a vibrational measurement. Additionally, vibrational spectra of heavy hydrocarbons become less sensitive to structural changes. The structural change in large-molecular-weight molecules results in relatively small or little spectral change compared to that in small-molecular-weight molecules. The application of vibrational techniques for the analysis of heavy petroleum products such as crude oil, pitch, and visbraker residue has been reviewed.[13] In those cases, vibrational techniques were used as supplemental analytical tools for in-depth investigation of heavy product characteristics, instead of direct measurement of bulk properties.

The determination of API (American Petroleum Institute) gravity (141.5 ÷ specific gravity − 131.5) of AR with the use of NIR spectroscopy has been studied in this paper. The API gravity is one of the most important properties that determine the operation conditions of distillation columns. Initially, other vibrational techniques such as IR and Raman were evaluated. Both techniques were found to be unsuitable for the analysis of AR due to fluorescence from polynuclear aromatic hydrocarbons in AR (Raman) and low reproducibility including signal-to-noise ratio (IR). The weakly absorbing nature and high incident radiation energy of NIR enabled transmission through AR samples with a reasonable pathlength, as well as the collection of highly reproducible NIR spectra. With the use of partial least-squares (PLS) regression, the necessary information was effectively extracted from NIR spectra and correlated to data from conventional methods.

EXPERIMENTAL

Sample Preparation. Eighty-one AR samples were collected over a four-month period at SK Corporation (Ulsan, Korea). Over this long period, samples were stra-

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tegically collected to present more process-related variations into the data set, which in turn increased the robustness of the calibration models. The variation of API gravity in the data set was from 12.1 to 19.2 (average: 15.7, standard deviation: 2.1). Immediately after collection, samples were sealed and stored at room temperature. The API gravity of each sample was measured with a hydrometer.[14]

Spectral Collection and Data Processing. NIR spectra were collected over the 1100 to 2500 nm spectral region with a NIRSystems Model 6500 spectrometer (Foss-NIRSystems Inc., Silver Spring, MD) equipped with a quartz halogen lamp and PbS detector. The spectra were collected with a rectangular quartz cuvette (pathlength: 0.5 mm) with a temperature control unit. Sample temperature can be controlled up to 60 °C (± 0.1 °C). The resolution of collected spectra was 10 nm with 2 nm data intervals. Spectra were collected by transferring heated AR samples into a cuvette and controlling the temperature at 60 °C. AR is a solid (similar to a candle) at room temperature; therefore, it requires heating in order to flow. Background spectra of air were collected for every sample immediately before the collection of the sample single-beam spectrum.

IR spectra (3500 to 600 cm^{-1}) were collected with a MIDAC 730 SX FT-IR spectrometer equipped with an attenuated total reflection (ATR) probe (ZnSe), KBr beamsplitter, and DTGS (deuterated triglycine sulfate) detector. Each spectrum was collected by positioning the ATR probe into each sample in a sealed bottle. Each IR spectrum corresponded to an accumulation of 128 scans with a resolution of 4 cm^{-1}. All IR spectra were collected at 60 °C.

Fourier transform (FT)-Raman spectra were collected on a Bruker IFS55 FT-IR spectrometer equipped with a Bruker FRA 106 Raman module, CaF_2 beamsplitter, and cryogenically cooled Ge detector. The excitation source was a diode pumped Nd:YAG laser operated at 1064 nm. The sample was transferred into a rectangular cuvette with back-reflecting mirror, and 128 scans were taken at a resolution of 4 cm^{-1} in a 180 ° scattering arrangement.

A total of 81 NIR spectra were divided into 51 spectra for calibration and 30 spectra for the prediction data set. The division of samples was intended to assign roughly two thirds of the total spectra into the calibration set. Spectra in the calibration and prediction set were chosen randomly. All data processing, including second derivative and PLS regression, was accomplished by using NSAS Ver. 3.3 software (Foss-NIRSystems Inc.).

RESULTS AND DISCUSSION

Vibrational Spectral Features (Raman, IR, and NIR). Initially, Raman, IR, and NIR spectra were collected in order to examine the spectral features of AR. Each vibrational spectrum is shown in Fig. 1. As shown in the figure, the Raman spectrum of AR is dominated by fluorescence despite a low excitation laser power (5 mW). With excitation power greater than 5 mW, the detector was completely saturated, and no further Raman signal was observed. The fluorescence is primarily due to the polycyclic aromatic hydrocarbons (asphaltenes) in AR. Asphaltenes are insoluble in normal hydrocarbon liquids such as *n*-hexane. Asphaltenes have a high average molecular weight (approximately 3600 A.U.) and a broad weight distribution.[15] Since asphaltenes have a highly conjugated structure, these result in strong fluorescence even with very low NIR laser power. To decrease the fluorescence, we diluted raw AR samples in *n*-heptane (4 g AR in 100 mL of *n*-heptane), and corresponding Raman spectra were collected with a laser power of 5 mW. As shown in Fig. 1, the fluorescence from diluted AR is decreased compared to that of raw AR. Additionally, hydrocarbon peaks (C–H stretching) around 2900 cm^{-1} overlapped with fluorescence are observed. Even with dilution, the strong fluorescence is still much more dominant than the vibrational features of AR. Therefore, Raman spectroscopy, even with low-power NIR excitation, cannot be used for further quantitative analysis, even

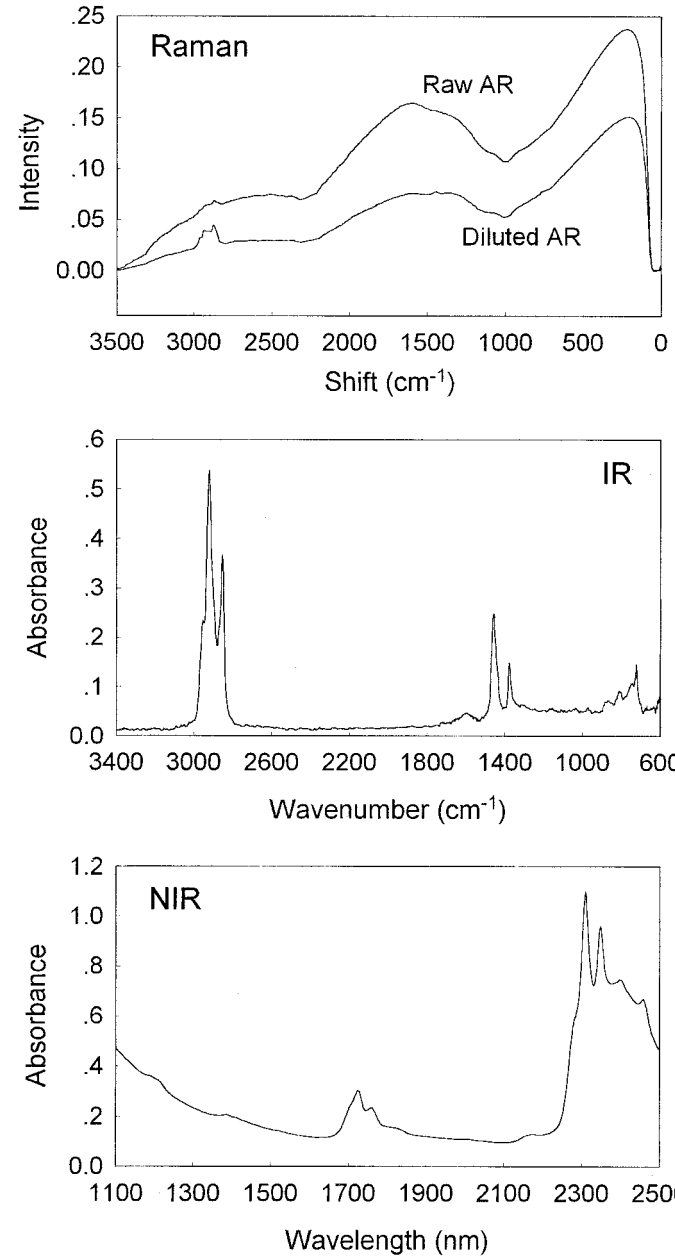

FIG. 1. Raman, IR, and NIR spectra of typical AR.

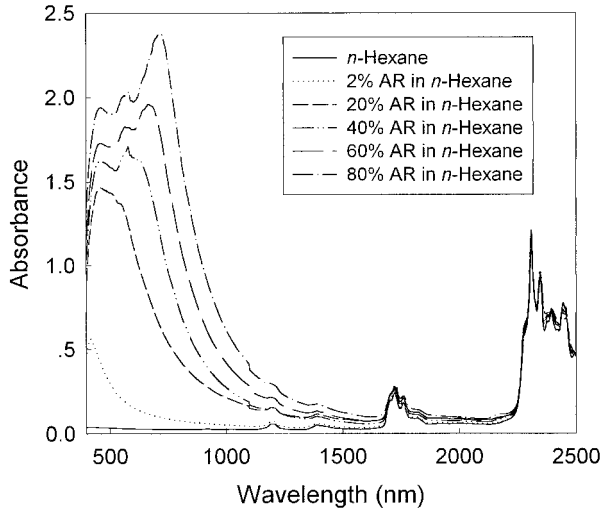

FIG. 2. Visible/NIR spectra of *n*-hexane and diluted AR samples in *n*-hexane.

though acquisition of a Raman spectrum is relatively simple and easy.

IR measurement of AR requires a very short pathlength for transmission due to dark and highly absorbing characteristics of the sample. In practical usage, control and use of a short pathlength for routine IR transmission measurement are less reproducible and cumbersome. Therefore, ATR was coupled with the IR technique. IR spectra of AR collected with an ATR geometry show reasonable spectral features of fundamental vibrations. The collection of an IR spectrum by transmission has been attempted; however, it was hard to control pathlength and obtain reproducible IR spectra in terms of peak shape, in comparison to ATR. The most useful information from IR is in the 3080–2750 and 1700–600 cm^{-1} regions. The bands at 3080–2750 and 1700–600 cm^{-1} correspond to C–H stretching and bending, respectively. However, the IR spectrum is too noisy for quantitative measurements (as can be seen in Fig. 1).

Conversely, an NIR spectrum of AR shows broad spectral features without the appearance of noise (as seen in the IR spectrum). Even though the spectral information in NIR was broad and highly overlapping, it contained relevant vibrational features with high reproducibility. The most useful information is in the 1650–2500 nm (6060–4000 cm^{-1}) region. The bands around 1700 nm correspond to the first overtone of C–H bands from the fundamental vibrations in the 3080–2750 cm^{-1} range, as shown in the IR spectrum. The absorption bands in the 2100–2500 nm region correspond to the combination bands, which are due to the coupling between the C–H stretching band in the 3080–2750 cm^{-1} region and the CH_3/CH_2 bending band around 1400 cm^{-1} of the IR. The baseline in the 1100–1650 nm range shows an exponential decay pattern. This feature is from the strong absorption in the visible region due to the dark color of AR. Figure 2 shows the visible/NIR spectra of *n*-hexane and *n*-hexane diluted AR samples. As expected, there is no significant visible absorption band of pure *n*-hexane. As the AR concentration is increased from 2 to 80%, the absorption band in the visible range is significantly increased; this result directly influences the bands in the NIR region, especially in the shorter wavelength NIR region (1100–16500 nm). No significant effect is observed in the longer wavelength NIR region (2100 to 2500 nm), which is farther from the visible absorption band. The color of 2% AR was light brown; however, 20 to 80% of the samples were completely black. The results indicate that when NIR is used for AR analysis, the baseline of NIR spectra will be influenced by the variation of AR quality from the different operating conditions of the crude distillation unit.

TABLE I. Average intensity, standard deviation, and relative standard deviation (RSD) of 20 spectra of the same AR sample in different spectral ranges for both IR and NIR.

	Spectral range	Average intensity (absorbance unit)	Standard deviation (absorbance unit)	RSD (%)
IR	3500–600 cm^{-1}	0.04878	0.0024	7.83
	3080–2750 & 1700–600 cm^{-1}	0.0809	0.0027	3.56
NIR	1100–2500 nm	0.2686	0.0010	0.46
	1650–2500 nm	0.2962	0.0008	0.35

Overall, IR provides the best qualitative features and resolution compared to NIR. Conversely, NIR yields broad spectral features with low resolution, but provides the highest signal-to-noise ratio over IR, even by simple visual examination.

Reproducibility Test of IR and NIR. It was previously determined that spectral reproducibility (including signal-to-noise ratio) of spectra directly influences the performance of a quantitative calibration developed by chemometric techniques.[16] Before quantitative chemometric analysis, the spectral reproducibility of both IR (ATR geometry) and NIR (transmission) measurements was examined. To do this, we continuously collected both IR and NIR spectra of the same AR sample (20 spectra) over an 8 h period at 30 min intervals on the same day. A background single-beam spectrum of air was collected for each measurement immediately before collection of the sample single beam spectrum. To evaluate spectral reproducibility, we calculated average absorbance and standard deviation of 20 spectra at each wavelength and calculated the corresponding relative standard deviations (RSDs) at each wavelength interval. The RSD was used to evaluate the reproducibility of both spectroscopic techniques. Then, those calculated values at each wavelength were averaged for different spectral ranges where absorption bands were present. The results are summarized in Table I. As shown in Table I, NIR shows the best reproducibility (10- to 17-fold) compared to that of IR. The average RSDs of NIR spectra in the spectral ranges containing absorption bands are superior to those from IR spectra.

The RSDs at each wavelength for IR and NIR are graphically shown in Fig. 3. Both plots in Fig. 3 compare RSDs from all 20 AR spectra. The AR IR spectra were offset for better qualitative comparison. RSD patterns for IR and NIR are similar. The lowest RSDs from both techniques were achieved in the hydrocarbon absorbing region. In the IR, hydrocarbon absorbing regions (3080–2760 and 1450 cm^{-1}) show lower RSDs compared to nonabsorbing regions (3400–3080 and 2760–1700 cm^{-1})

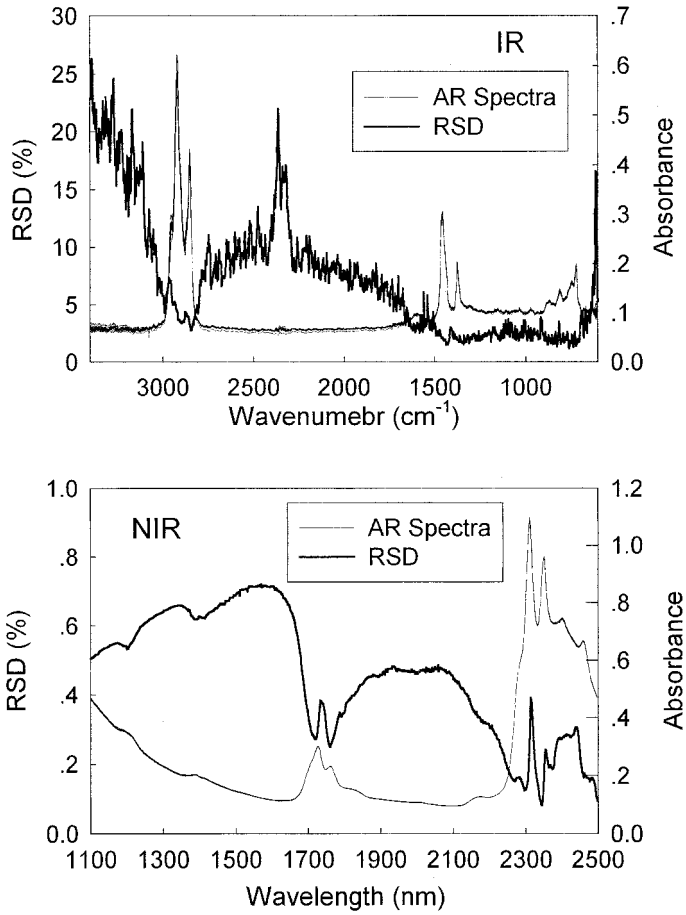

FIG. 3. The relative standard deviations (RSDs) at each wavelength for IR (top) and NIR (bottom). Both figures are compared with all 20 AR spectra.

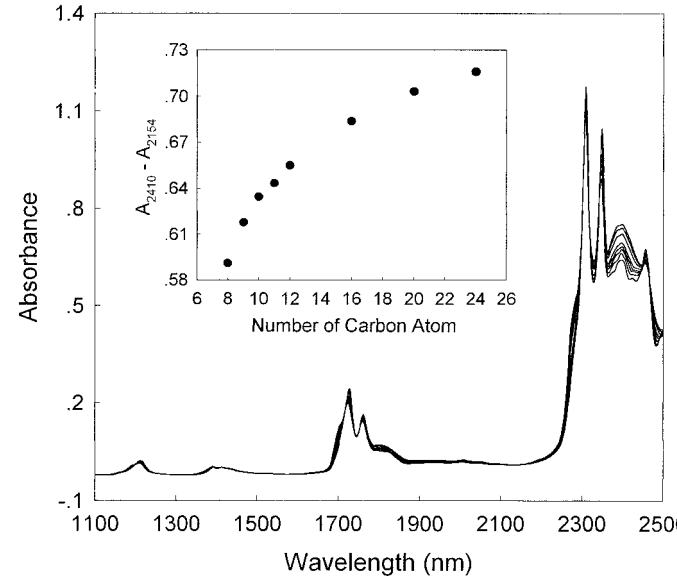

FIG. 4. NIR spectra of normal linear hydrocarbons with different chain lengths [from bottom to top: octane (C_8), nonane (C_9), decane (C_{10}), undecane (C_{11}), dodecane (C_{12}), hexadecane (C_{16}), eicosane (C_{20}), and tetracosane (C_{24})]. The correlation between number of carbon atoms and difference in absorbance at 2410 and 2154 nm is plotted inside the figure.

which contain noise. The observed noise is a result of a low signal-to-noise ratio. Similarly, the NIR RSDs around 1650–1900 and 1800–2200 nm regions are nearly the same; however, weakly absorbing regions at 1100–1650 and 1900–2100 nm ranges exhibit higher RSDs. It is difficult to make absolute signal-to-noise comparisons of a dispersive method (NIR) to a Fourier transform method (IR) since FT is a multiplexed method that distributes noise across all wavelengths. However, it is obvious that NIR exhibits the higher signal-to-noise ratio over IR as based on Fig. 3.

Consequently, NIR was finally selected for the analysis of AR because it provided more reproducible spectra with higher signal-to-noise ratio than IR or Raman. Even though bands in the NIR are broad and less characterized, the necessary information related to a desired property can be extracted by using multivariate calibration methods such as PLS regression.

NIR Spectral Sensitivity Based on the Number of Carbon Atoms. To apply NIR spectroscopy to the analysis of heavy petroleum products, one must initially examine the spectral sensitivity of NIR because absorption bands in the NIR are inherently weak and spectral variation based on structural change will be less sensitive as the hydrocarbon chain length increases. For the analysis of heavy hydrocarbons by NIR, small spectral variations are required as the chain length increases. For this purpose, NIR spectra of normal linear hydrocarbons [octane (C_8), nonane (C_9), decane (C_{10}), undecane (C_{11}), dodecane (C_{12}), hexadecane (C_{16}), eicosane (C_{20}), and tetracosane (C_{24})] were collected to examine the spectral sensitivity as a function of hydrocarbon chain length. All the spectra were collected at 60 °C. Figure 4 shows NIR spectra of normal linear hydrocarbons with different chain lengths. As expected, the spectral variations become less apparent as the number of carbon atoms increases. To systematically investigate the spectral variations, we plotted the correlation between number of carbon atoms and difference in absorbance at 2410 and 2154 nm. As shown in Fig. 4, the absorbance at 2410 nm is increasing sharply up to a carbon atom number of 12, and then gradually increases up to a carbon atom number of 24. As discussed previously, NIR spectra of higher molecular weight hydrocarbons become less sensitive in terms of their structural changes since the structural change in large molecules results in relatively small spectral variation in comparison to that in low-molecular-weight molecule. Tetracosane ($C_{24}H_{50}$, m.p.: 56 °C, b.p.: 391 °C) can be regarded as a lower molecular weight compound among components of AR. On the basis of Fig. 4, when the carbon number is more than 24, there are reasonable spectral variations even though the spectral variations become less sensitive.

Overall, the absorption bands in NIR are broad and less characteristic; however, spectral variations are observed in higher molecular weight hydrocarbons. These spectral variations can provide quantitative information that is useful for calibration when using PLS regression. Additionally, NIR spectra provide excellent reproducibility, which also improves the quantitative calibration performance, as has been previously studied.

PLS Calibration Models. Figure 5 shows all 51 raw (original absorbance) and second-derivative spectra of

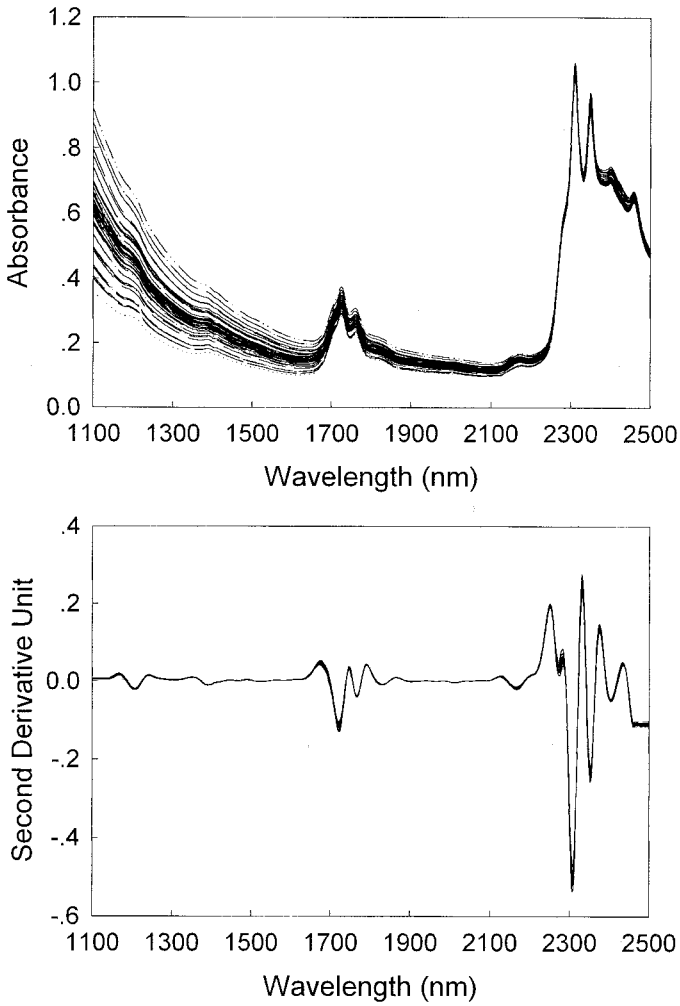

FIG. 5. All 51 raw (top) and second-derivative spectra (bottom) of AR in the calibration data set.

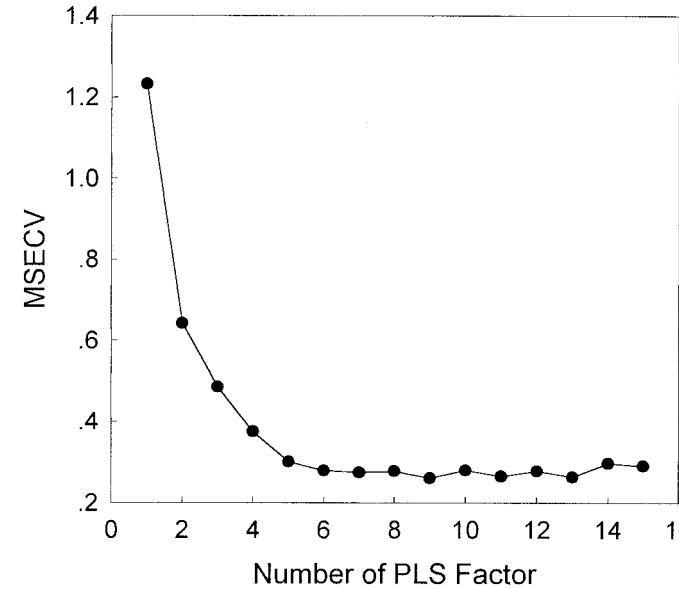

FIG. 6. MSECV plotted as a function of the number of PLS factors used to determine the API gravity within the 2100–2500 nm range.

AR that were used in the calibration data set. In the raw NIR spectra, as discussed previously, baselines vary significantly due to strong visible absorption, especially in the 1100–1650 nm range. The degree of baseline variation decreases at longer wavelengths. It is well known that second derivatives enhance the spectral features and remove (or at least reduce) baseline variation. As shown in Fig. 5, the spectral variations are effectively enhanced while keeping the same qualitative information, while removing the baseline variations. The most noticeable spectral variations are observed in the 1650–1900 and 2100–2500 nm ranges.

In the application of a PLS algorithm, it is generally known that the spectral range and number of PLS factors are critical parameters.[7] Four different spectral ranges (1100–2500, 1650–2500, 1650–1900, and 2100–2500 nm) were examined in this study. The first range utilizes the whole NIR spectrum. The second range includes both first-overtone (around 1700 nm) and combination bands (around 2300 nm) and excludes the 1100–1650 nm range where the baseline variations are large. The third and fourth spectral ranges include only overtone and combination regions, respectively.

The number factor where the MSECV (mean square error of cross-validation) value reaches a minimum can be identified as an optimum; however, it can result in overfitting, especially for predicting unknown samples. To eliminate overfitting problem, we identified the optimum number of factors by applying the F-test to MSECV values. The number of factors where F-test probability level falls to or below 0.75 was chosen as optimum in this study. Figure 6 shows the MSECV plotted as a function of the number of PLS factors that were used to determine the API gravity within the 2100–2500 nm range. The leave-one-out cross-validation was used as a validation method. The MSECV decreases sharply for the initial factors and then gradually decreases as related spectral variation is incorporated into the calibration model. By examining only MSECV values (minimum MSECV value), we chose nine PLS factors as the optimum; however six PLS factors were selected by applying the F-test. As shown in the plot, MSECV values do not significantly decrease after six factors. Therefore, the model using six factors is less likely to be overfit compared to that using nine factors. The optimum number of PLS factors for other spectral regions was determined by applying the F-test to MSECV values.

The results of the PLS calibration employing both raw and second-derivative spectra for each spectral range are summarized in Table II. Overall, the use of second-derivative spectra helps to improve calibration slightly. The best result was achieved by using the second-derivative spectra in 1650–2500 nm range, which excluded the baseline varying region (1100–1650 nm), and simultaneously including both first-overtone (1650–1900 nm) and combination regions (2100–2500 nm). When the full spectral range that included the baseline varying region was utilized, the calibration results were worse. The baseline variation (which is not related to API gravity) slightly affects the calibration. Using only small spectral regions (first-overtone or combination bands) results in relatively worse calibration results. Overall, a wider spectral range that does not contain the baseline varying region

TABLE II. PLS calibration results in different spectral ranges employing both raw (original absorbance) and second-derivative spectra.

Spectra range (nm)	Number of PLS factors	MSECV	SEP
Raw spectra			
1100–2500	7	0.30	0.27
1650–2500	6	0.30	0.28
1650–1900	6	0.29	0.29
2100–2500	6	0.29	0.31
Second-derivative spectra			
1100–2500	4	0.30	0.27
1650–2500	6	0.27	0.22
1650–1900	6	0.28	0.28
2100–2500	7	0.29	0.26

results in better calibration models, by incorporating more spectral information.

A scatter plot showing the correlation between NIR and the reference ASTM method for API gravity of AR, using the second-derivative spectra in the 1650–2500 nm range, is shown in Fig. 7. Open and filled circles represent the calibration and prediction data, respectively. The calibration and prediction data have good correlation to the conventional reference data, and many points fall on or close to the unity line. API gravities in the prediction data set were accurately predicted with a standard error of prediction (SEP) of 0.22.

When NIR spectroscopy is applied to actual process control, the repeatability becomes an important parameter that has to be considered. If the measurement has unacceptable repeatability, the process control will deviate from a control target and thus minimize operation efficiency. Conversely, with highly repeatable analytical data, maximum operation is possible by effectively utilizing the process equipment up to its limit. To examine the repeatability of NIR, we continuously collected a spectrum of the same AR sample in 30 min intervals over approximately 2 days (45 h, 90 spectra collected). The reference spectrum of air was collected at each measurement, and temperature was controlled at 60 °C during the entire measurement time. Then, in each measurement, API gravity was predicted by using the calibration model developed with the use of the second-derivative spectra in the 1650–2500 nm range. The results are shown in Fig. 8 in comparison to the ASTM repeatability. Solid, dotted, and dashed lines correspond to average, standard deviation of NIR measurement, and ASTM repeatability, respectively. The average and standard deviations of predicted API gravity were 18.27 and 0.038, respectively. NIR provides approximately three times better repeatability compared to ASTM repeatability, which is 0.1. The excellent instrumental stability and sufficient penetration of NIR radiation through AR provide highly repeatable analytical data. The superior repeatability of NIR will significantly improve the process control performance compared to using a conventional on-line analyzer.

CONCLUSION

The overall results in this study present the possibility of using NIR spectroscopy for the analysis of heavy petroleum products, such as AR and crude oil. So far, applications of vibrational spectroscopy in this field have not been pioneered to a great extent; however, we have demonstrated the feasibility of further expansion of vibrational spectroscopic applications to heavy petroleum products. Raman and IR spectroscopy were not appropriate analytical methods for the analysis of AR due to strong fluorescence and poor reproducibility, respectively. Even though baselines were varying (due to strong visible absorption of dark samples), weakly absorbing and high radiation energy of NIR penetrated the sample, and highly reproducible spectra were collected. The high reproducibility of NIR leads to the development of accurate calibration models using PLS regression. API gravities in the prediction data set were accurately predicted with an SEP of 0.22. Additionally, NIR provided

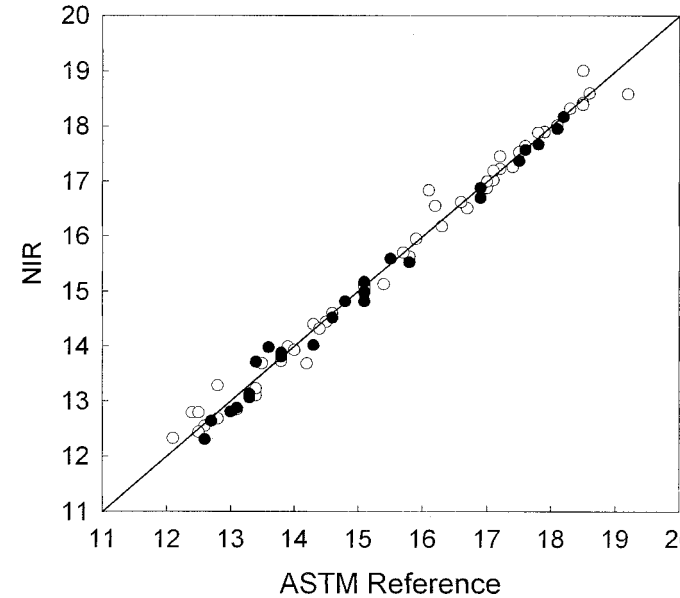

FIG. 7. Scatter plot showing the correlation between NIR and reference ASTM method for the calibration model utilizing second-derivative spectra in the 1650–2500 nm range. Open and filled circles correspond to calibration and prediction data, respectively.

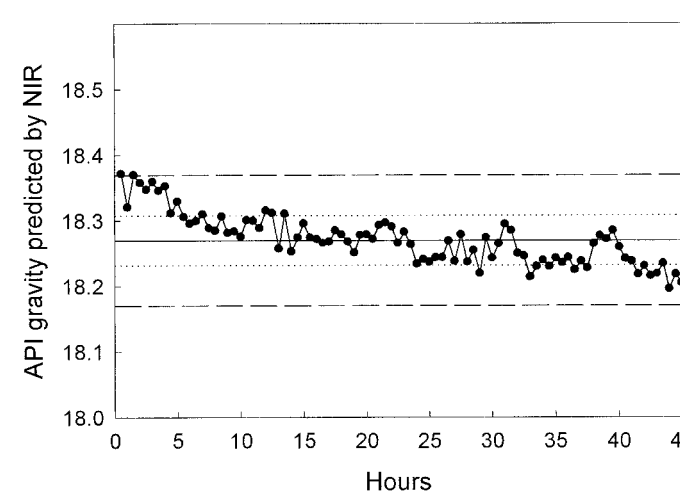

FIG. 8. The repeatability of NIR measurements. Solid, dotted, and dashed lines correspond to average, standard deviation of NIR measurements, and ASTM repeatability, respectively. NIR spectra of the same AR sample were collected over 45 h at 30 min intervals.

highly repeatable analytical data, which can be perfectly suited for tight process control and optimization. Future studies will examine NIR spectroscopy for determination of other physical properties of AR such as viscosity and distillation temperature, and evaluate an on-line process NIR analyzer system for real-time process control.

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